GIORGIO MORRA

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US Work Status: F-1 Student ***Eligible for OPT***

INTRODUCTION

Mechanical Engineering graduate student from Politecnico di Milano specializing in Mechatronics and Robotics, currently conducting Master's Thesis research at NASA JPL. I aim to contribute to innovative solutions for the control and navigation of Autonomous Systems.

EDUCATION

M.Sc. in Mechanical Engineering (Mechatronics and Robotics) 2023 – Ongoing
Politecnico di Milano, Italy
International Mobility Extra-EU: Pontificia Universidad Javeriana, Colombia

B.Sc. in Mechanical Engineering 2020 – 2023
Politecnico di Milano, Italy
Erasmus+ Program: Universidad Pontificia Comillas, Spain

EXPERIENCE

Robotics Navigation & State Estimation Intern *Dec 2025 – Ongoing*
NASA Jet Propulsion Laboratory – Robotic Mobility and Manipulation Section Pasadena, CA

Contributed to the **Venus Variable Altitude Aerobot** project, focusing on localization strategies for the Venusian atmosphere.

- Modeled the balloon's flight dynamic and an interplanetary (Venus-Earth-Sun) communication model;
- Simulated onboard measurements for sun sensors, radar altimeters, and accelerometers;
- Designed the state prediction model and selected optimal filtering techniques for trajectory reconstruction;
- Performed sensitivity analyses to assess state estimation accuracy under varying sensor availability and noise levels;
- Developed a high-fidelity thermal FEM model to simulate the thermodynamics of the external balloon.

UNIVERSITY PROJECTS

UAV Autonomous Navigation in Urban Environment

- Autonomy Architecture: Engineered a navigation pipeline from 3D global routing to real-time trajectory optimization for autonomous vehicles;
- Path Planning and Trajectory Optimization: Implemented an A* global path planner with shortcutting and cubic-spline smoothing to generate time-parameterized reference trajectories;
- Non-Linear MPC Optimization: Developed a receding-horizon controller utilizing CasADi/IPOPT to execute precise trajectory tracking under strict speed and acceleration constraints;
- Dynamic Obstacle Avoidance: Integrated reactive collision avoidance constraints within the MPC, enabling the vehicle to safely bypass moving targets unknown to the global planner.

1-DoF Robotic Exoskeleton for Elbow Rehabilitation

- Developed a 1-DoF collaborative elbow exoskeleton to support flexion-extension rehabilitation for post-stroke patients;

- Implemented a ”teach-and-play” control strategy, a force sensor, and a servo motor to automate therapeutic motions;
- Designed admittance control for transparent trajectory recording and passive control for accurate, autonomous movement playback;
- Modeled system kinematics and dynamics, successfully validating requirements for motion smoothness, tracking precision, and user safety.

Design, Simulation, and Dynamic Modeling of a 4-DOF SCARA Robot

- Developed a comprehensive kinematic and dynamic simulation model for a SCARA industrial robotic arm;
- Mapped the operational workspace and analyzed manipulability ellipses to evaluate the robot's kinematic dexterity and performance;
- Designed and optimized motion planning algorithms, comparing the computational efficiency and smoothness of trajectories generated via lines, parabolas, and splines;
- Formulated a multi-body dynamic model to accurately estimate joint torques under varying payload conditions.

SKILLS

Prog. Languages: Python, Matlab, C++
Simulation Tools: Simulink, Simscape, ROS2
CAD Software: Inventor, SolidWorks, Solid Edge, CATIA V5
Various: LaTeX, Microsoft Office

LANGUAGES

Italian (Native) — English (C1) — Spanish (C1)